BioSift

Predicting Molecular Bioactivity Using Physicochemical Descriptors and Machine Learning

Prepared by: Judi Yousri
Student ID: 231002400
Course: CBIO313 — Data Mining & Machine Learning

1. Project Overview

BioSift is a machine learning powered web application designed to predict the bioactivity potential of small molecules against cancer cell lines. The system uses physicochemical molecular descriptors as input features and classifies compounds as Active or Inactive using a Random Forest classifier trained on data from the National Cancer Institute (NCI) bioassay screening program.

The project covers the complete machine learning lifecycle: data ingestion and cleaning, exploratory data analysis, molecular descriptor engineering using the RDKit cheminformatics library, multi-model training and evaluation, and deployment as an interactive Streamlit web application.

The motivating idea behind BioSift is straightforward. Drug discovery is a slow and costly process, with an average cost of approximately 2.6 billion US dollars per approved drug. High-throughput screening programmes such as the NCI bioassay generate large volumes of experimental data, but manually analysing this data to identify promising compounds does not scale. BioSift explores whether a machine learning model can learn to predict bioactivity directly from a small set of physicochemical descriptors grounded in Lipinski's Rule of 5 — the framework medicinal chemists have used for decades to assess drug-likeness — providing a fast, accessible first-pass screening tool before any laboratory work is undertaken.

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2. Dataset

2.1 Source

The dataset originates from the National Cancer Institute (NCI) PubChem BioAssay database, containing high-throughput screening results of small molecule compounds tested against human cancer cell lines.

2.2 Structure

- 58,219 rows remaining after removal of metadata header rows
- 18 original columns, including SMILES structure strings, activity outcomes, and GI50/TGI growth-inhibition measurements
- Binary classification target: Active (1) versus Inactive (0)

2.3 Class Distribution

The dataset exhibits severe class imbalance, which became the central technical challenge addressed throughout the project:

- Inactive: 51,641 compounds (93.1%)
- Active: 3,821 compounds (6.9%)

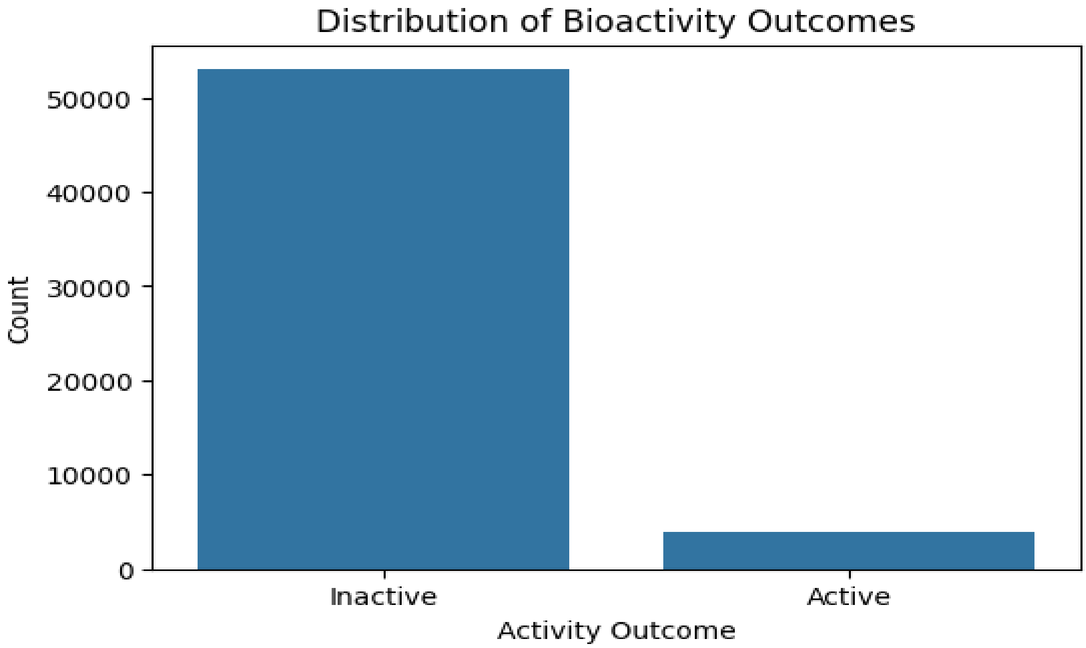

Figure 1 — Distribution of Bioactivity Outcomes (Active vs Inactive)

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3. Data Preprocessing

3.1 Cleaning Steps Applied

- Removed the top two metadata rows of the raw Excel export (result type and description headers)
- Converted seven numerical columns from object dtype to numeric using pd.to_numeric(errors='coerce')
- Dropped low-value columns: PUBCHEM_ACTIVITY_URL and PUBCHEM_ASSAYDATA_COMMENT
- Dropped four columns with greater than 99% missing values: LogGI50_u, LogGI50_V, LogTGI_u, LogTGI_V
- Applied median imputation to remaining missing numerical values to preserve sample size
- Removed rows lacking SMILES structure strings, as these are required for descriptor computation
- Filtered LogP to a realistic physicochemical range of [-5, 8], removing erroneous outlier values as extreme as 41.8

After all cleaning steps, the working dataset contained 55,462 usable compound records.

3.2 Activity Score and Potency Distributions

The Activity Score distribution confirms the class imbalance visually, with the overwhelming majority of compounds clustering near zero. The LogGI50_M distribution represents the logarithm of the concentration required to inhibit 50% of cancer cell growth; more negative values indicate more potent compounds. Most screened compounds in this dataset require micromolar concentrations, considered relatively weak activity, while values below -7 are generally regarded as genuinely potent hits.

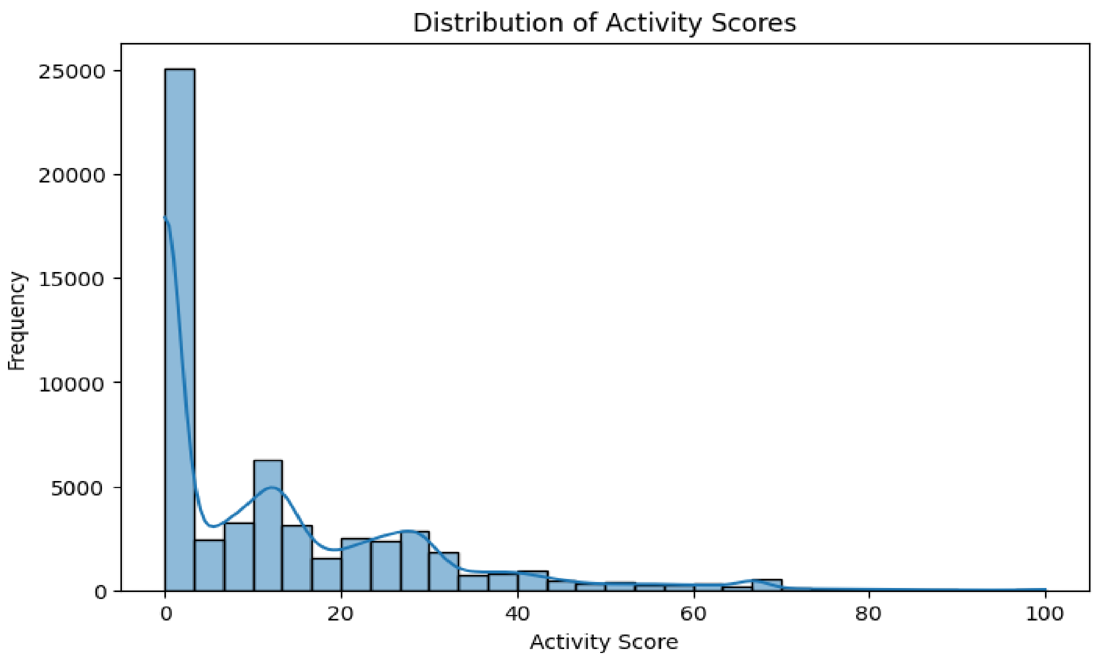

Figure 2 — Distribution of Activity Scores across the dataset

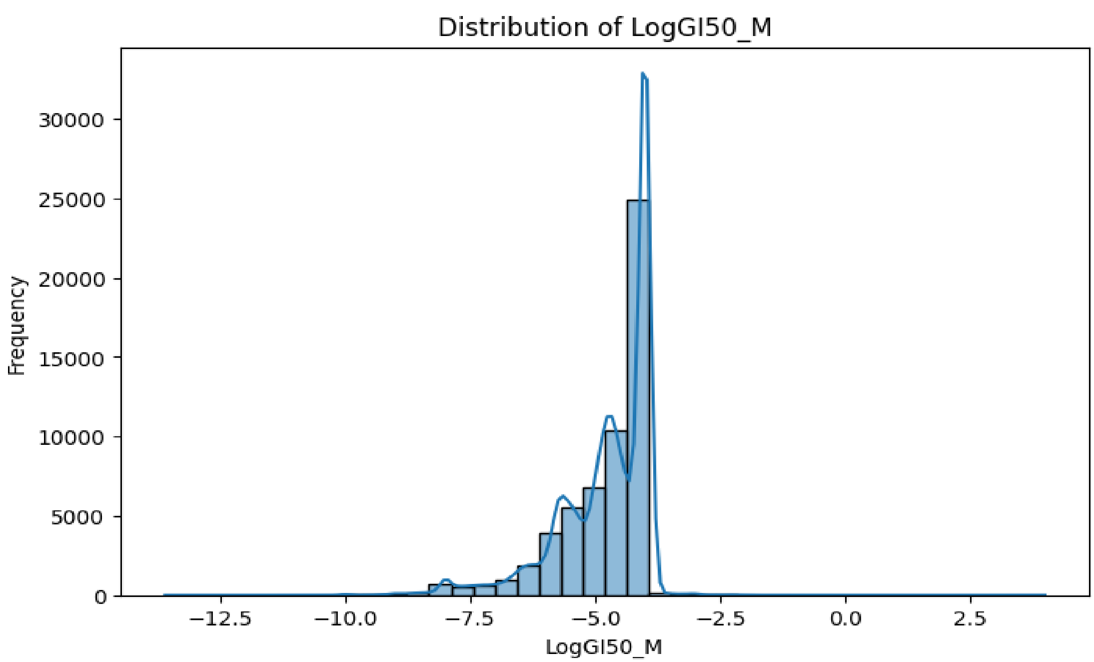

Figure 3 — Distribution of LogGI50_M values

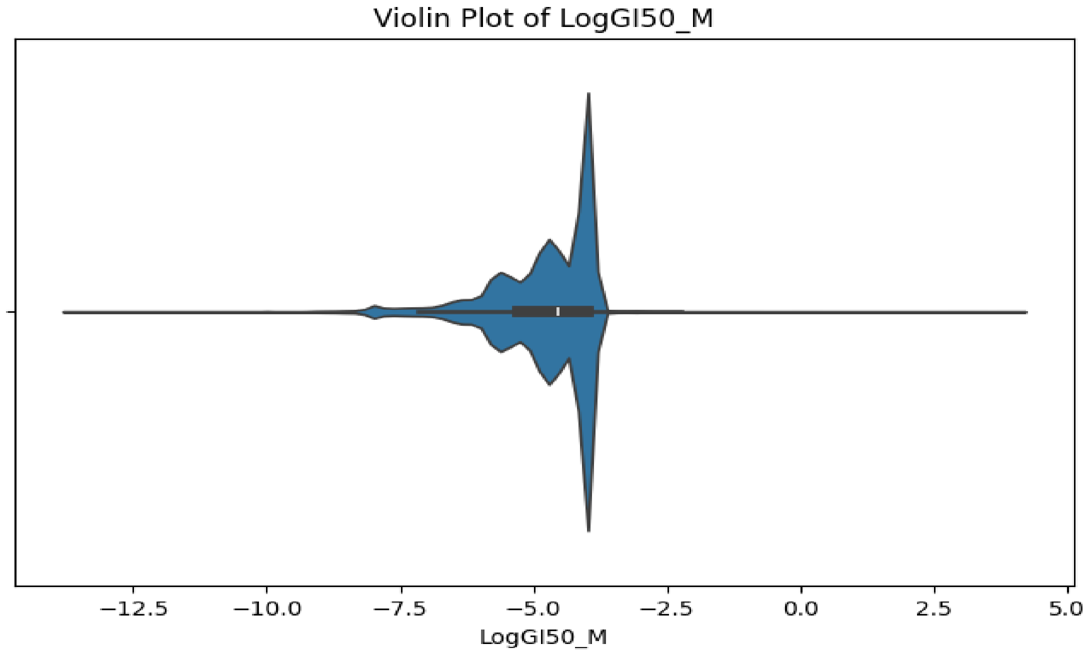

Figure 4 — Violin Plot of LogGI50_M showing spread and density

3.3 Correlation Among Raw Bioassay Measurements

A correlation heatmap of the raw bioassay measurements was generated to identify redundant variables. LogGI50_M and LogTGI_M show a strong positive correlation, which is biologically expected since both measure growth inhibition at different endpoints — GI50 corresponds to 50% inhibition and TGI to total growth inhibition. The Activity Score also correlates moderately with GI50, confirming that the NCI-assigned score is a meaningful summary of compound potency rather than an arbitrary label.

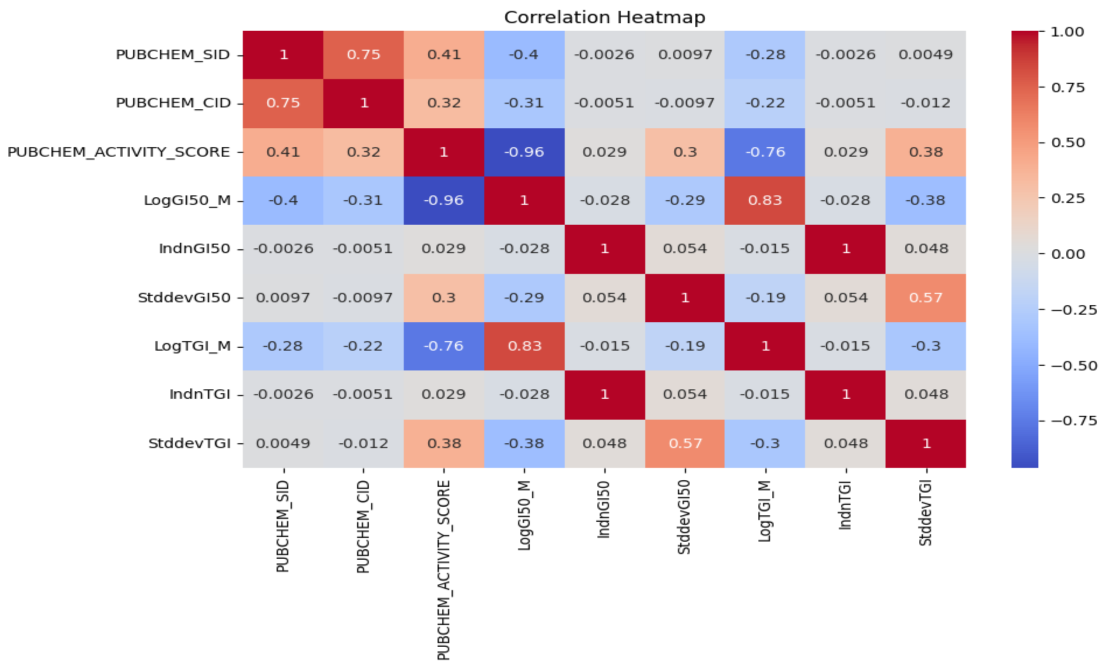

Figure 5 — Correlation Heatmap of raw bioassay measurements

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4. Molecular Descriptor Engineering

4.1 RDKit Descriptor Computation

Molecular descriptors were computed directly from SMILES structure strings using the RDKit cheminformatics library. The five descriptors selected for the final feature set correspond to the parameters defined by Lipinski's Rule of 5, a framework developed by Christopher Lipinski at Pfizer in 1997 that remains the standard heuristic for assessing oral drug-likeness.

4.2 Lipinski's Rule of 5 — Feature Definitions

Descriptor	Lipinski Threshold	Rationale
Molecular Weight (MolWt)	<= 500 Da	Large molecules are poorly absorbed through the intestinal wall
LogP	<= 5	Very lipophilic compounds accumulate in fat tissue and resist dissolution
H-Bond Donors	<= 5	Excess donor groups increase polarity, hindering membrane crossing
H-Bond Acceptors	<= 10	High acceptor counts raise the energetic cost of desolvation

TPSA	<= 140 A^2	Large polar surface area reduces passive membrane permeability

4.3 Descriptor Distributions and Relationships

The Molecular Weight distribution is right-skewed, indicating a small number of very large molecules in the dataset. The LogP boxplot comparing Active and Inactive outcomes is particularly informative: Active compounds tend to exhibit lower LogP values than Inactive compounds, suggesting that moderate polarity is associated with bioactivity against these cancer cell lines. The descriptor correlation matrix shows that Molecular Weight, H-Bond Donors, H-Bond Acceptors, and TPSA are all inter-correlated, since larger molecules tend to carry more polar functional groups; despite this collinearity, all five descriptors were retained as each contributed uniquely to model performance.

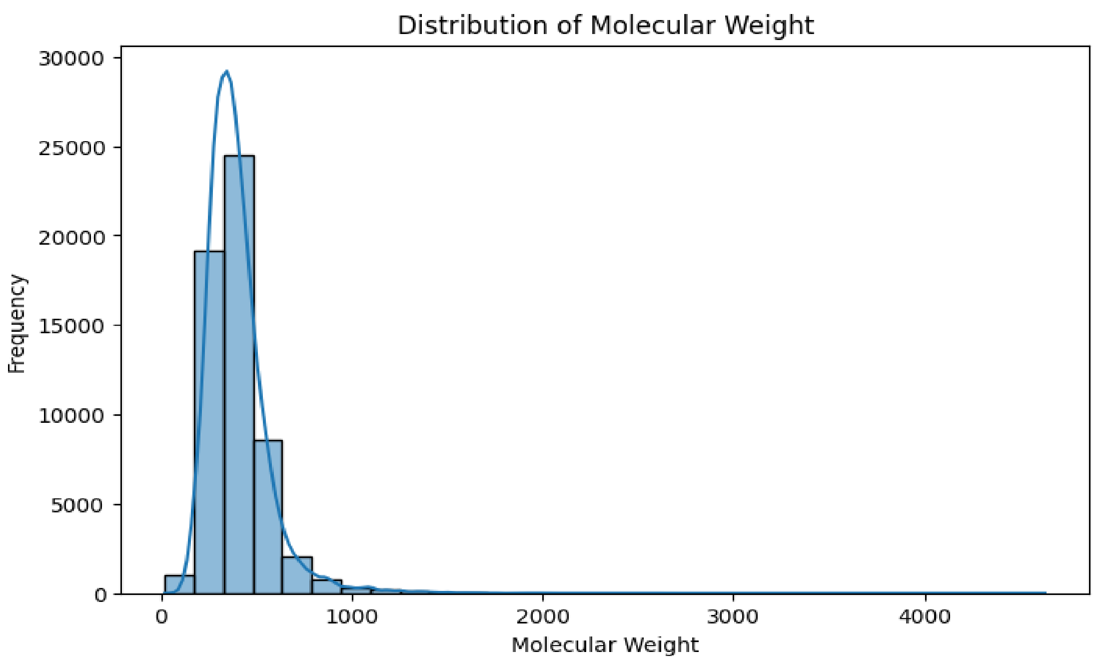

Figure 6 — Distribution of Molecular Weight across screened compounds

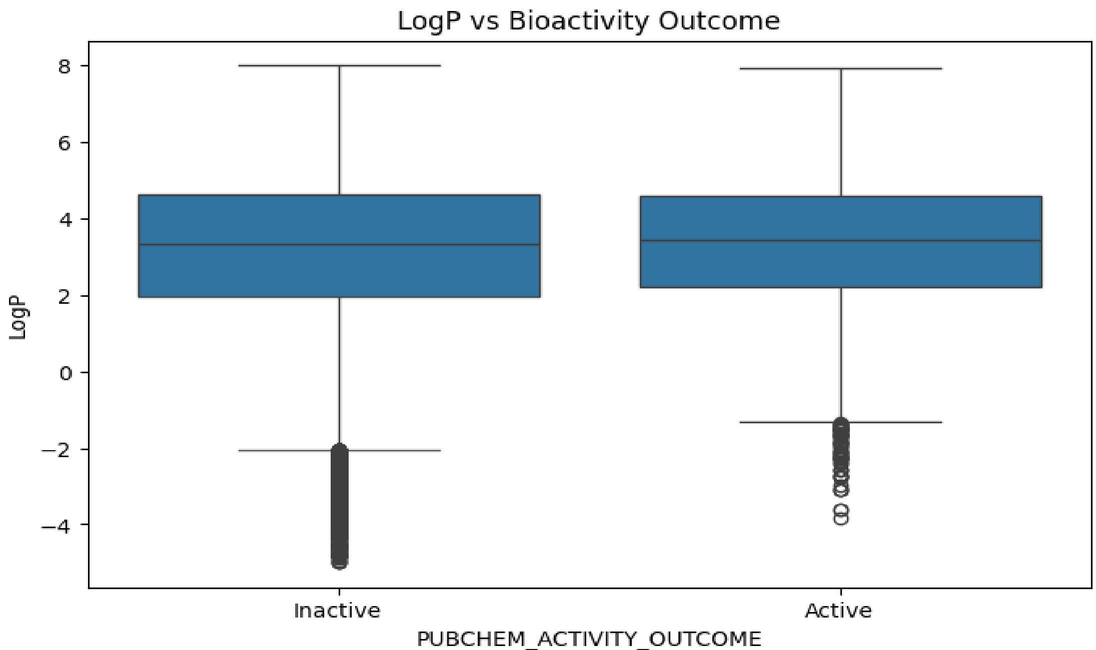

Figure 7 — LogP distribution compared across Active and Inactive outcomes

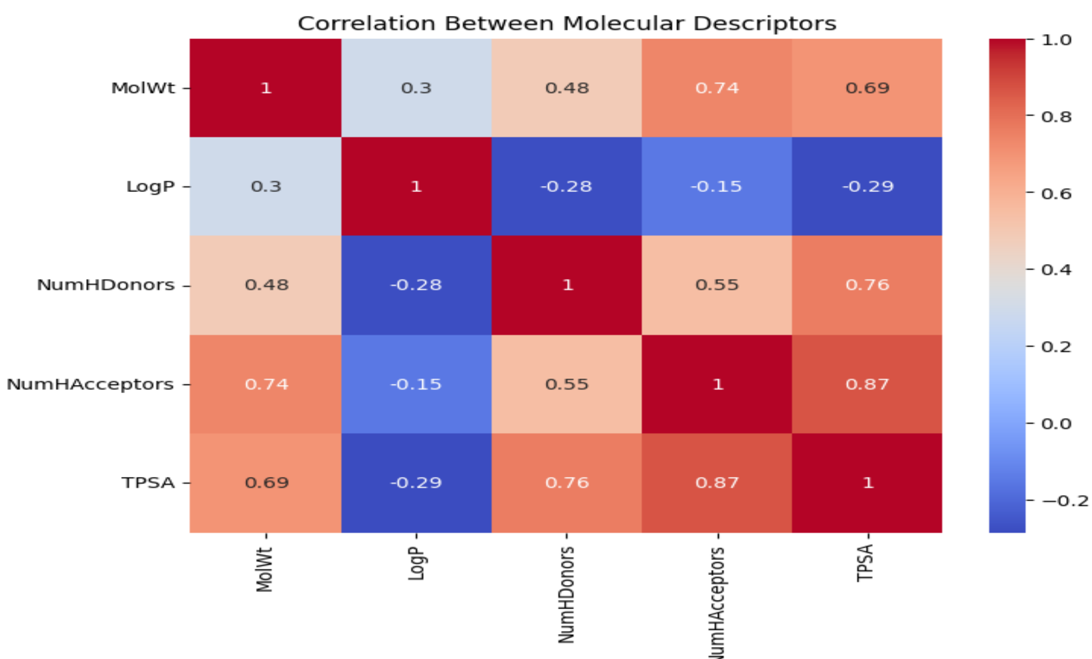

Figure 8 — Correlation matrix of the five molecular descriptors

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5. Model Training and Evaluation

5.1 Train-Test Split

The dataset was partitioned into 80% training (44,369 samples) and 20% testing (11,093 samples) subsets using stratified sampling, preserving the 93:7 class ratio in both partitions.

5.2 Models Evaluated

- Logistic Regression — linear baseline trained on standardized features
- Random Forest — ensemble of 200 decision trees; selected as the final production model
- XGBoost — gradient boosting framework
- Support Vector Machine (SVM) — kernel-based classifier
- K-Nearest Neighbors (KNN) — instance-based classifier

5.3 Handling Class Imbalance

Given the severe 93:7 imbalance between Inactive and Active classes, the final Random Forest model was trained with class_weight='balanced', which automatically assigns the Active class approximately 13 times more weight during training. This single change was the most consequential decision in the project — without it, the model degenerates into a classifier that predicts Inactive for almost every input, an outcome that achieves high accuracy (93.1%) while providing zero practical value as a screening tool.

5.4 Model Comparison — Accuracy and F1-Score

All five models achieve above 92% accuracy, but this metric is largely uninformative given the class imbalance: a model that always predicts Inactive would itself score 93.1%. The F1-score for the Active class is the metric that meaningfully differentiates the models, and Random Forest leads by a wide margin — at 45.3% it is more than 15 percentage points ahead of the next-best model (KNN at 29.0%), and nearly ten times higher than the weakest model (Logistic Regression at 4.6%).

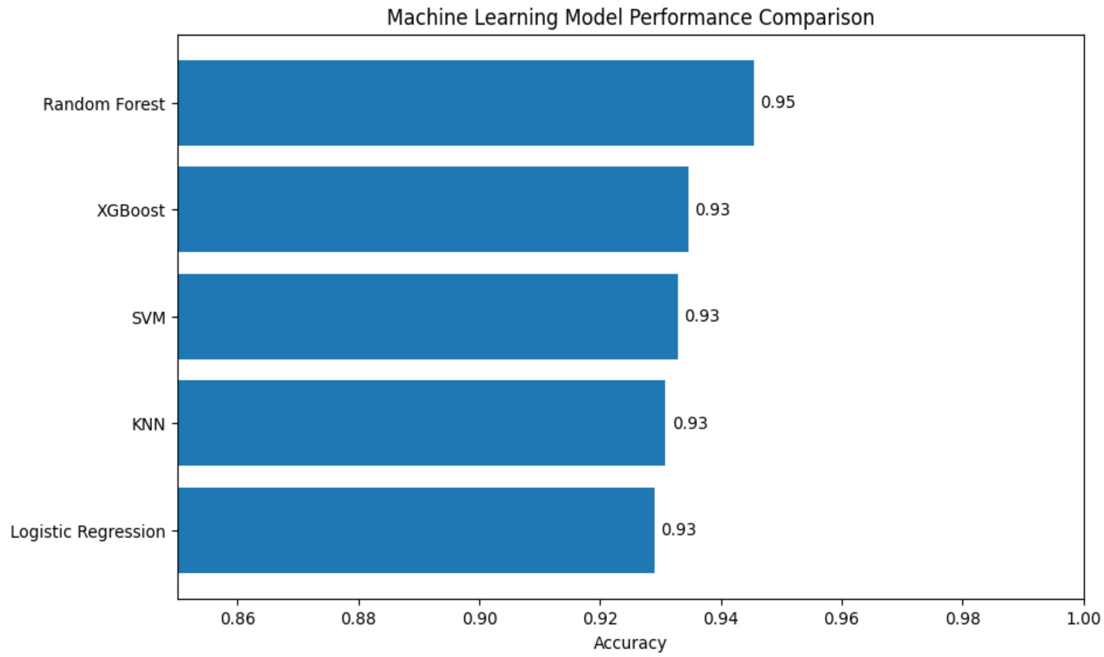

Figure 9 — Model accuracy comparison across all five algorithms

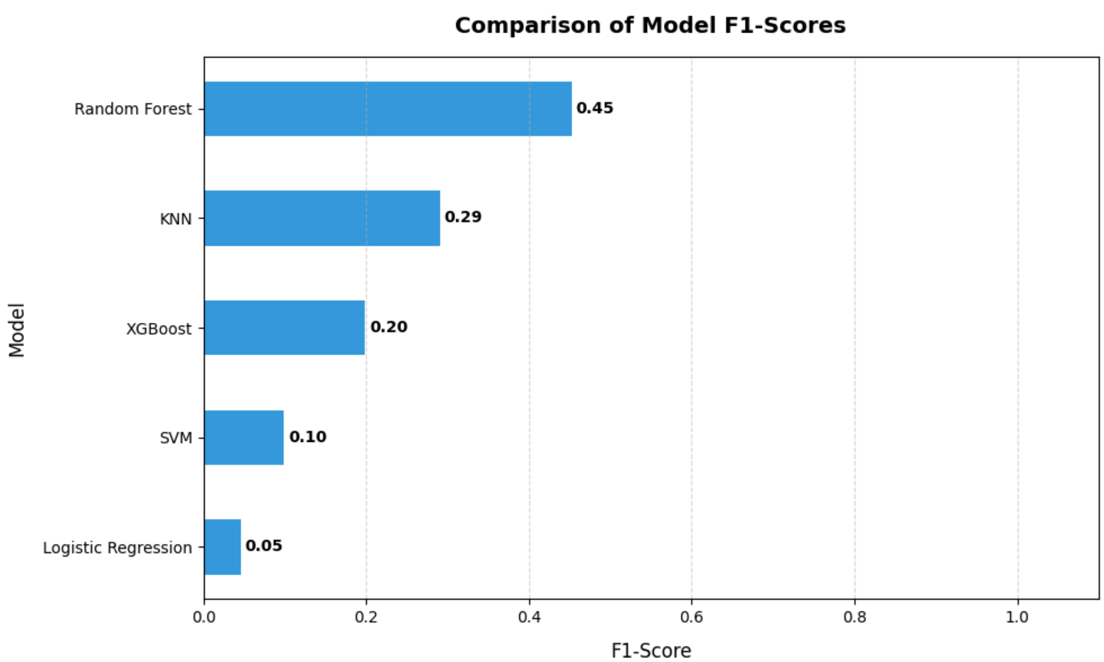

Figure 10 — F1-Score comparison across all five algorithms

5.5 Full Evaluation Metrics

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	94.6%	73.5%	32.7%	45.3%
KNN	93.1%	49.5%	20.5%	29.0%
XGBoost	93.5%	63.8%	11.8%	19.9%
SVM	93.3%	66.1%	5.4%	9.9%
Logistic Regression	92.9%	30.6%	2.5%	4.6%

Random Forest achieved the highest score across all four evaluation metrics and was selected as the production model. Its recall for the Active class (32.7% at the default 0.50 probability threshold) reflects the inherent difficulty of the class imbalance problem; the deployed BioSift application applies a lowered classification threshold of 0.15 to improve detection of active compounds, trading some precision for substantially improved recall.

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6. Feature Importance

Random Forest provides built-in feature importance scores derived from the mean decrease in impurity across all 200 decision trees. The resulting ranking reveals which molecular properties are most predictive of bioactivity in this dataset:

- Molecular Weight — 30.1% (dominant predictor)
- Topological Polar Surface Area (TPSA) — 26.8%
- LogP (lipophilicity) — 25.7%
- H-Bond Acceptors — 11.1%
- H-Bond Donors — 6.2%

Molecular Weight, TPSA, and LogP together account for 82% of the model's predictive power, validating the choice of descriptor set: the model has effectively learned to weight molecular properties in a manner consistent with established medicinal chemistry heuristics. The relatively low contribution of H-Bond Donors (6.2%) is notable; the Lipinski donor threshold is the rule most commonly violated by known active natural products, which may partly explain its weaker discriminative power in this dataset.

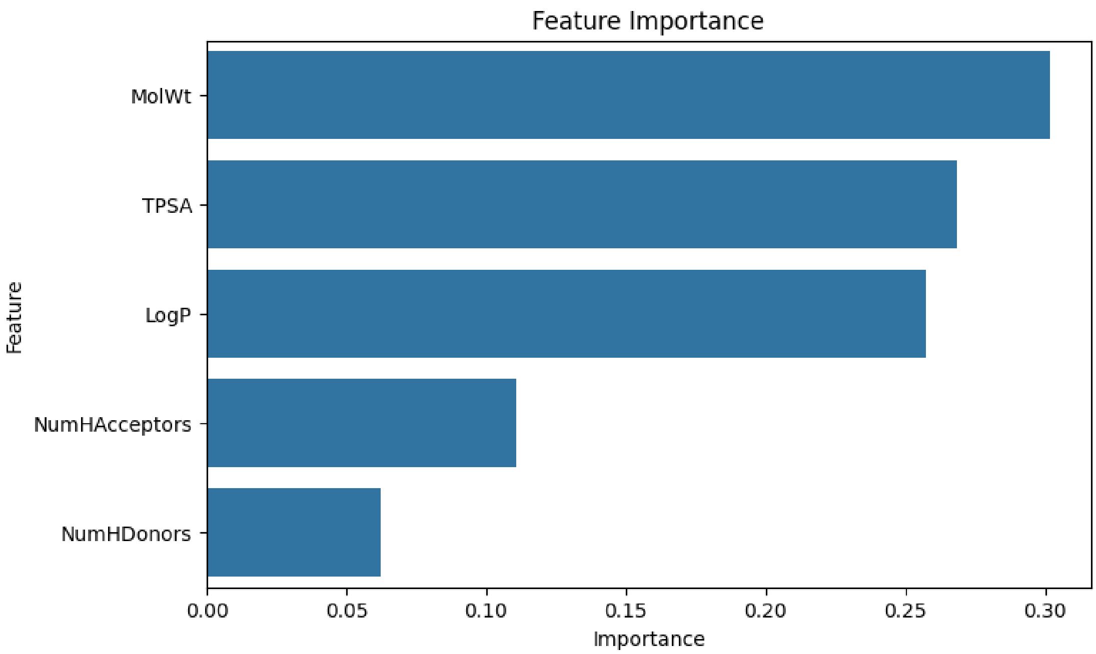

Figure 11 — Feature Importance scores from the Random Forest model

7. Confusion Matrix and ROC Curve

7.1 Confusion Matrix

The confusion matrix on the held-out test set illustrates the model's prediction breakdown. The model correctly identifies the large majority of Inactive compounds, reflected in a high true negative rate, while the comparatively lower true positive rate for Active compounds visualises the recall limitation discussed in Section 5.

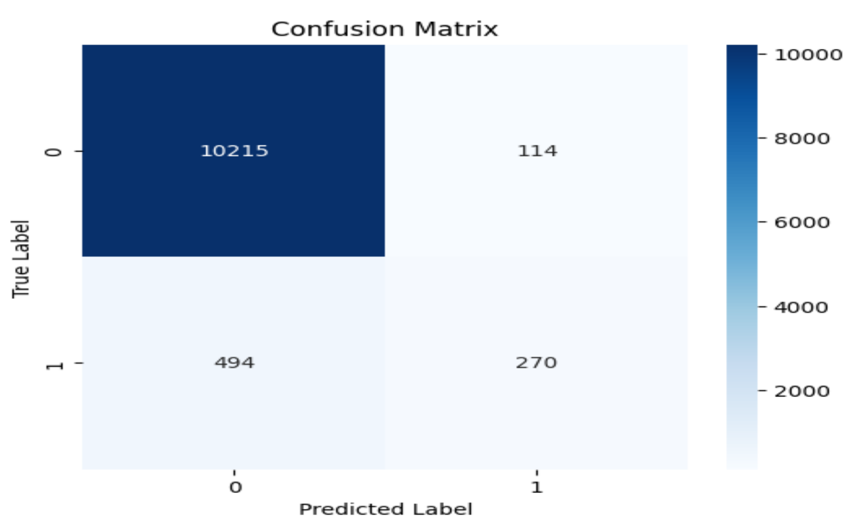

Figure 12 — Confusion Matrix on the held-out test set

7.2 ROC Curve

The ROC curve, generated using XGBoost predictions, plots the true positive rate against the false positive rate across all possible classification thresholds. The resulting Area Under the Curve (AUC) score indicates that the model's probability outputs are well-ordered — active compounds consistently receive higher predicted probabilities than inactive compounds — which is the underlying justification for lowering the classification threshold in the deployed application to recover additional true positives.

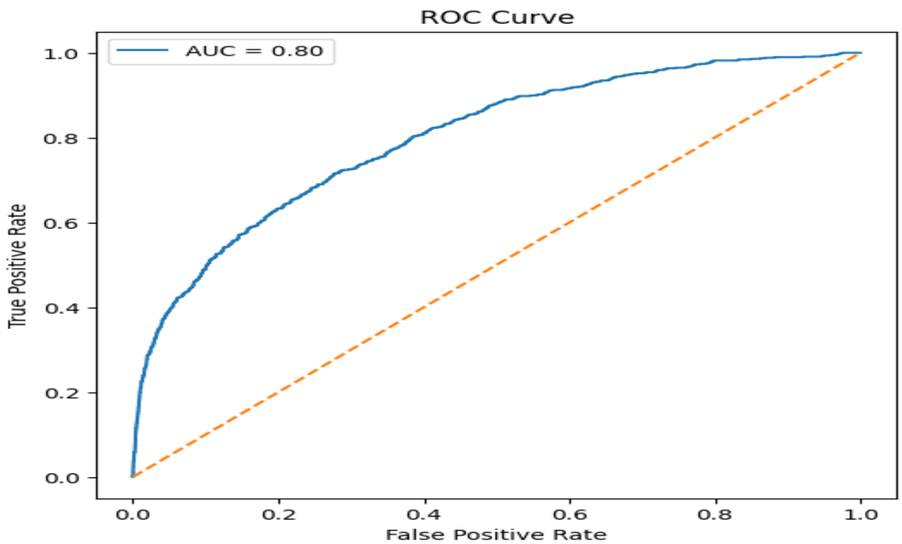

Figure 13 — ROC Curve with AUC score (XGBoost)

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8. The BioSift Web Application

8.1 Overview

BioSift is a Streamlit-based front-end application that wraps the trained Random Forest model in an interactive user interface. It allows users to input molecular descriptor values and receive real-time Active/Inactive predictions, accompanied by class probability scores and compound-specific biochemical insights.

8.2 Key Features

- Landing page presenting the project description and a notice regarding ongoing dataset expansion
- User Control tab — five descriptor input fields (Molecular Weight, LogP, H-Bond Donors, H-Bond Acceptors, TPSA)
- User Guide tab — documentation of Lipinski's Rule of 5 and descriptor explanations

- Six pre-loaded trial compounds drawn from the NCI dataset (three Active, two Inactive, and one Manual Entry option)
- Real-time Active/Inactive classification with associated probability percentages
- Biochemical Insight Note panel providing compound-specific structural and pharmacological context
- Scroll-triggered animated indicator directing users to insights following a screening run
- Fully adaptive dark mode and light mode styling, detected automatically via system preference
- Low-opacity molecular bond structure background graphic

8.3 Technology Stack

- Python 3.12
- Streamlit 1.58.0
- scikit-learn — RandomForestClassifier
- RDKit — molecular descriptor computation from SMILES
- pandas and numpy — data processing
- joblib — model serialization

8.4 Model Artifacts and Deployment

- bioactivity_model.pkl — trained Random Forest (200 estimators, class_weight='balanced')
- features.pkl — ordered list of the five descriptor feature names

The model was trained on raw, unscaled descriptor values; no StandardScaler is applied, and the application passes raw descriptor inputs directly to the model. The application is deployed via GitHub to Streamlit Community Cloud, with Git Large File Storage (LFS) used to host the 120 MB model file, which exceeds GitHub's standard 100 MB repository file size limit.

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9. Pre-loaded Trial Compounds

Six trial compounds were selected directly from the NCI dataset to demonstrate the model's behaviour across a representative range of outcomes.

Compound	Outcome	NCI Score	MolWt	LogP	Donors	Acceptors	TPSA
A-1 (CID 44415057)	Active	100/100	1155.46	0.85	9	24	342.0
A-2 (CID 6197)	Active	55/100	253.34	0.62	3	4	83.9
A-3 (CID 2154)	Active	61/100	441.40	-1.85	5	10	206.8
I-1 (CID 11122)	Inactive	10/100	122.12	0.95	0	2	34.1
I-2 (CID 219128)	Inactive	15/100	364.27	5.32	0	0	0.0

Compound A-1 is a depsipeptide macrocyclic antibiotic scaffold, the highest-scoring compound in this trial set at 100/100. Its molecular weight exceeds 1,100 Da and it possesses 24 hydrogen bond acceptors, allowing it to wrap tightly around target proteins and lock enzymatic function — similar scaffolds underpin approved antibiotics such as daptomycin.

Compound A-2 is a cyclohexanone-lactam natural product (MW 253) that satisfies all five Lipinski criteria, with a low TPSA permitting passive membrane diffusion and a moderate LogP balancing solubility and permeability.

Compound A-3 is structurally related to Pemetrexed, an FDA-approved chemotherapy agent for lung cancer and mesothelioma. Its strongly negative LogP (-1.85) requires active transport into cancer cells via folate receptors, which are overexpressed in tumour tissue.

Compound I-1 is a 2,6-dimethyl-1,4-benzoquinone fragment. At only 122 Da, it is too small to engage meaningfully with any receptor binding pocket, despite quinone scaffolds appearing in bioactive natural products such as Vitamin K.

Compound I-2 is an iodoalkyl triphenyl lipophilic scaffold with a LogP of 5.32 and a TPSA of exactly 0 A^2. With zero hydrogen bond donors or acceptors, it cannot form any of the stabilising interactions required to bind selectively to a protein target, and instead accumulates non-specifically in lipid membranes.

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10. Limitations and Future Work

10.1 Current Limitations

- Only five physicochemical descriptors are used; structurally distinct compounds can share identical descriptor profiles, limiting discriminative power
- Active compound recall remains at approximately 35%, meaning the model misses roughly two out of three genuinely active compounds
- The dataset is limited to NCI cancer cell line assays and may not generalise to other biological targets
- No three-dimensional structural information, molecular fingerprints, or graph-based representations are incorporated

10.2 Proposed Improvements

- Incorporate extended fingerprint features (Morgan / ECFP4 fingerprints) for richer structural representation at the atomic level
- Apply SMOTE or related oversampling techniques during training to further address class imbalance
- Evaluate Graph Neural Network (GNN) architectures operating directly on molecular bond-connectivity graphs
- Expand the training dataset with additional NCI assays and external ChEMBL compound data
- Integrate SHAP explainability visualisations into the BioSift interface to support interpretation of individual predictions

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11. Conclusion

This project developed an end-to-end machine learning pipeline for small molecule bioactivity prediction. Beginning with raw NCI bioassay data, the system performs comprehensive data cleaning, computes RDKit molecular descriptors grounded in Lipinski's Rule of 5, trains and evaluates five classification algorithms, and deploys the best-performing model — a Random Forest achieving 94.6% accuracy — through the BioSift web application.

The central technical challenge throughout the project was the severe class imbalance present in the NCI dataset, with Inactive compounds outnumbering Active compounds by approximately 13 to 1. This was addressed through class-weighted training and a lowered classification threshold in the deployed application. While the five-descriptor feature set captures the key pharmacokinetic properties defined by Lipinski's Rule of 5 and achieves strong overall performance, the limited recall for Active compounds highlights clear directions for future work involving richer structural representations such as molecular fingerprints and graph neural networks.

BioSift demonstrates that meaningful drug-likeness prediction can be achieved from simple, interpretable physicochemical descriptors alone, providing an accessible and practical tool for early-stage computational screening in drug discovery research.